

3. Generative Adversarial Networks (GANs)

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Previous Week

- ▶ Autoencoder
- ▶ VAE

Historical Impact of GANs

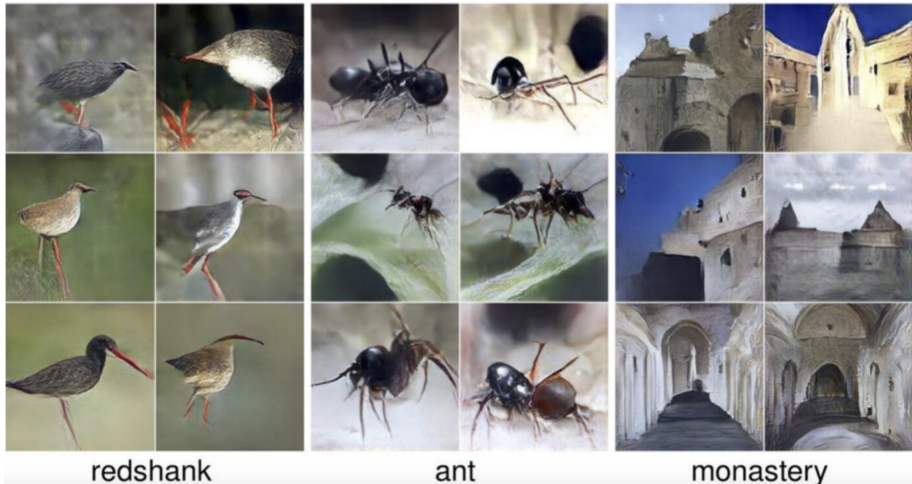
- ▶ GAN was introduced by Ian Goodfellow.
- ▶ Before GANs: Generative models produce blurry images, over-smoothed outputs, but are computationally expensive.
- ▶ Since 2014, GANs could produce more realistic images.
- ▶ Later, StyleGAN could generate photorealistic human faces.
- ▶ Between 2018 and 2020, StyleGAN dominated image generation.
- ▶ Triggered rapid progress in Generative AI.
- ▶ Huge impact on computer vision and multimedia synthesis.

GAN Generated Images 1 (2016)



volcano

GAN Generated Images 2 (2016)



StyleGAN Generated Images (2019)



Fake or Real?



What is GAN?

- ▶ Core idea: train two neural networks against each other.
- ▶ Generator: Creates fake data.
- ▶ Discriminator: Detects fake data.
- ▶ Through competition, the fake examples become more realistic
- ▶ Analogy: Counterfeiter vs police
- ▶ Better police → better counterfeiter
- ▶ Better counterfeiter → better police

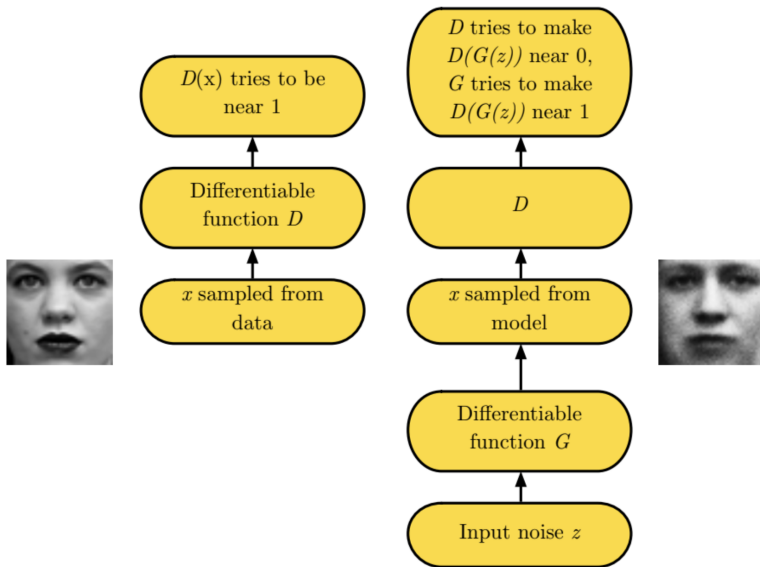
Learning for Generator

- ▶ Generator takes random noise.
- ▶ Over time, generator learns structures and patterns.
- ▶ The learning signal is from the discriminator.
- ▶ No explicit rules to learn.

Learning for Discriminator

- ▶ Receives both real examples and fake examples.
- ▶ Learns to distinguish between the two.
- ▶ The learning signal is from the classification error.
- ▶ Becomes a learned "critic" of realism.

GAN Framework



GAN Objective

- ▶ $D(x) = 1$: definitely real.
- ▶ $D(x) = 0$: definitely fake.
- ▶ $\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log (1 - D(G(\mathbf{z})))]$.
- ▶ Discriminator: maximise $V(D, G)$.
- ▶ Generator: minimise $V(D, G)$.
- ▶ When $D(x) = 1$ and $D(G(z)) = 0$, $V(D, G)$ maximised.
- ▶ When $D(x) = 0$ and $D(G(z)) = 1$, $V(D, G)$ minimised.
- ▶ The loss function of GAN is $-V(D, G)$.

Game Theory

- ▶ Multiple agents make decisions that affect each other
- ▶ Examples: Chess players, competing companies.
- ▶ Agents often have conflicting objectives.
- ▶ Classic paradox: prisoner's dilemma.
- ▶ Zero-sum game: gain of one player is the loss of the other.

Nash Equilibrium

- ▶ GAN is a 2-player zero-sum game.
- ▶ Both players try to improve continuously.
- ▶ Neither player can improve by changing their strategy alone.
- ▶ Best generator perfectly imitates real data: $p_g(x) = p_{\text{data}}(x)$.
- ▶ Best discriminator can only guess equally: $D(x) = 0.5$.
- ▶ Theoretical goal of GAN training.

Jensen-Shannon (JS) Divergence

- ▶ GAN wants $p_g(x) \approx p_{\text{data}}(x)$.
- ▶ To quantify the distribution difference, GAN uses JS divergence.
- ▶ Let $M(x) = \frac{1}{2}(p_{\text{data}}(x) + p_g(x))$.
- ▶ $JS(p_{\text{data}}||p_g) = \frac{1}{2}KL(p_{\text{data}}||M) + \frac{1}{2}KL(p_g||M)$.
- ▶ JS divergence is symmetric.
- ▶ $0 \leq JS(p_{\text{data}}||p_g) \leq \log 2$.
- ▶ If there is little overlap between $p_{\text{data}}(x)$ and $p_g(x)$, JS won't explode.

Jensen-Shannon (JS) Divergence Calculation

- ▶ $p_{\text{data}}(x)$ is unknown.
- ▶ $p_g(x)$ is unknown.
- ▶ GAN does not compute JS divergence directly.
- ▶ Discriminator $D(x) = P(\text{real}|x)$.
- ▶ Generator fixed, optimal $D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}$.
- ▶ $V(D^*, G) = -\log 4 + 2JS(p_{\text{data}}||p_g)$
- ▶ Discriminator acts as a surrogate to JS divergence.

GAN Applications

GAN Applications

- ▶ Artwork creation
- ▶ Fashion design
- ▶ Image translation: style transfer
- ▶ Super-resolution: medical images, satellite images
- ▶ Photo restoration and editing
- ▶ Audio generation
- ▶ Video generation
- ▶ Molecule design

Fake News



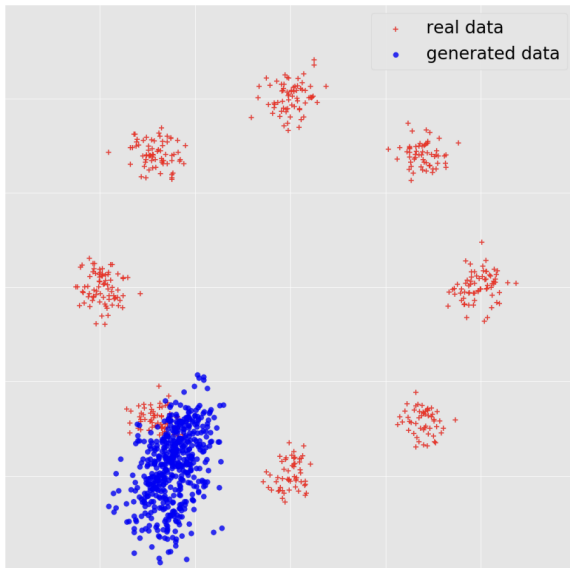
Deepfake Ethics

- ▶ Synthetic media imitating faces, voices, videos.
- ▶ Positive side: film entertainment, historical, educational.
- ▶ Ethical concerns: fake news, celebrity images, non-consensual synthetic media.
- ▶ Deepfake detection systems.
- ▶ Watermarking techniques.
- ▶ Disclosure of AI-generated content.

GAN Challenges

- ▶ Difficult adversarial training: optimisation target changes continuously.
- ▶ Vanishing gradients: discriminator is too strong too early.
- ▶ Mode collapse: limited variety of generated samples.

Mode Collapse Illustration



CNNs in GANs

- ▶ Convolutional layers preserve spatial information.
- ▶ Share parameters across locations.
- ▶ Learn hierarchical features.
- ▶ Common to have CNNs in both generator and discriminator.

DCGAN Generator

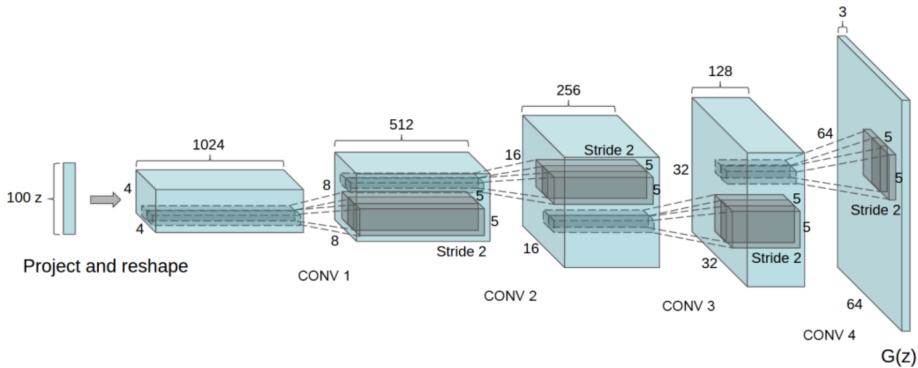


Image Quality Measurement

- ▶ Realistic
- ▶ Diverse
- ▶ Representative of the real data
- ▶ Popular metric: Fréchet Inception Distance (FID).
- ▶ FID extracts features from real images.
- ▶ FID measures the difference between extracted and generated features.
- ▶ Lower FID is better.

Summary

- ▶ GAN consists of a generator and a discriminator.
- ▶ GAN training can be viewed as a 2-player minmax game.
- ▶ GAN aims to reach the Nash equilibrium.
- ▶ GAN minimises JS divergence.
- ▶ GAN applications: image, audio, video.
- ▶ Deepfake ethics.
- ▶ GAN challenges: instability, vanishing gradients, mode collapse.
- ▶ Image generation evaluation.